DongJun **Lee**

Machine Learning & Optimization

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Education

Georgia Institute of Technology — *Atlanta, USA*
Ph.D. Student, Industrial Engineering — *Aug. 2025 - Aug. 2029*
- NSF AI4OPT Institute; Advisor: Prof. Pascal Van Hentenryck; Current GPA: 4.00 / 4.00

Seoul National University — *Seoul, South Korea*
M.S. in Artificial Intelligence — *Sep. 2023 - Aug. 2025*
- Data Science & AI Laboratory (DSAIL); Advisor: Prof. Sungroh Yoon; GPA: 4.25 / 4.30

Seoul National University — *Seoul, South Korea*
B.S. in Mechanical Engineering — *Mar. 2017 - Aug. 2023*
- Summa Cum Laude; GPA: 4.03 / 4.30

Research Experience

Research focus: Temporal learning and learning-augmented optimization for constrained, multi-period decision-making

Georgia Tech | NSF AI4OPT Institute — *Atlanta, USA*
Graduate Research Assistant ; Advisor: Prof. Pascal Van Hentenryck — *Sep. 2025 - Present*
- Investigating neural solution prediction and structured repair for large-scale, multi-period mixed-integer optimization with temporal and cross-resource constraints.
- Studying self-supervised proxies for multi-period resource-allocation linear programs using interval propagation and projection-based feasibility repair.
- Exploring context-conditioned decision policies for multistage stochastic optimization.

Seoul National University | DSAIL — *Seoul, South Korea*
Graduate Research Assistant; Advisor: Prof. Sungroh Yoon — *Sep. 2023 - Aug. 2025*
- Studied deep learning for long-range time-series forecasting and multivariate anomaly detection.
- Investigated temporal representations, spectral attention, and contrastive learning for long-range dependencies and multivariate structure.

Selected Publications

Under Review

2026 — W. Li, **D. Lee**, P. Van Hentenryck, "SMOP: Self-supervised Multi-period Optimization Proxy for Resource Allocation with Temporal Constraints," ***ICLR 2027*** (under review).

Published

2025 — H. Kim, J. Mok, **D. Lee**, et al., "Causality-Aware Contrastive Learning for Robust Multivariate Time-Series Anomaly Detection," ***ICML***.

2025 — J. Kim*, H. Kim*, H. Kim, **D. Lee**, S. Yoon, "A Comprehensive Survey of Deep Learning for Time Series Forecasting: Architectural Diversity and Open Challenges," ***Artificial Intelligence Review***. (* Equal Contribution)

2024 — B. G. Kang*, **D. Lee***, et al., "Introducing Spectral Attention for Long-Range Dependency in Time Series Forecasting," ***NeurIPS***. (* Equal Contribution)

Applied Research & Industry Collaborations

Explainable Attention-Based Credit Risk Modeling *Atlanta, USA*
Equifax | Industry Collaboration *Jan. - Apr. 2026*

- Built an attention-based credit risk modeling framework, including model tuning and aligned export of internal embeddings for analysis.
- Developed PCA/CCA-guided analyses connecting Transformer features with interpretable credit attributes.

Risk-Constrained Trading Agent *Atlanta, USA*
Independent Research *Aug. 2025 - Jan. 2026*

- Developed a multi-asset data pipeline and multi-task Transformer for ATR-scaled bracket-hit probabilities, evaluated with walk-forward validation.
- Implemented cost- and slippage-aware backtests with volatility-scaled sizing, exposure controls, and drawdown limits.

Learning-Based Drivetrain Simulator Calibration *Seoul, South Korea*
Hyundai | Industry Collaboration *Mar. - Jul. 2025*

- Developed learning-based calibration to align MATLAB drivetrain simulation outputs with real vehicle data, as an initial step toward ML-based transmission-parameter optimization proxy.

EV Battery Defect Anomaly Detection *Seoul, South Korea*
LG Energy Solution | Industry Collaboration *Aug. 2024 - Jun. 2025*

- Developed process-aware tokenization and learned representations for battery defect detection under class imbalance and noisy labels.
- Built a multimodal model combining time-series embeddings with expert-engineered tabular features.

ML-Based Motor Calibration *Seoul, South Korea*
Hyundai Genesis | Industry Collaboration *Mar. - Oct. 2024*

- Developed deep-learning methods to predict motor calibration maps from sparse operating-point data for reducing sixth-harmonic torque ripple and noise.

Internships

PricewaterhouseCoopers Consulting *Hwaseong, South Korea*
Associate Consultant Intern *Jul. 2023*

- Supported a consulting project with Logic, DRAM, and Flash research teams at a semiconductor company. (NDA)

OnePredict *Seoul, South Korea*
Data Scientist Intern *Jun. - Jul. 2021*

- Developed DL methods for turbine-system fault diagnosis using vibration and operational data.

Technical Skills

Programming: Python (PyTorch), Julia.

Optimization: Mathematical optimization, optimization proxies, feasibility repair, multi-period mixed-integer optimization, and multistage stochastic optimization.

Machine learning: Temporal representation learning, time-series forecasting, anomaly detection, and contrastive learning.

Earlier Publication

2016 **D. Lee**, T. Yoon, C. Lee, "A Study on the Posture Based Control of Robotic Fixation Device," ***International Journal of Applied Engineering Research***, Vol. 11, No. 9, pp. 6655–6659.